

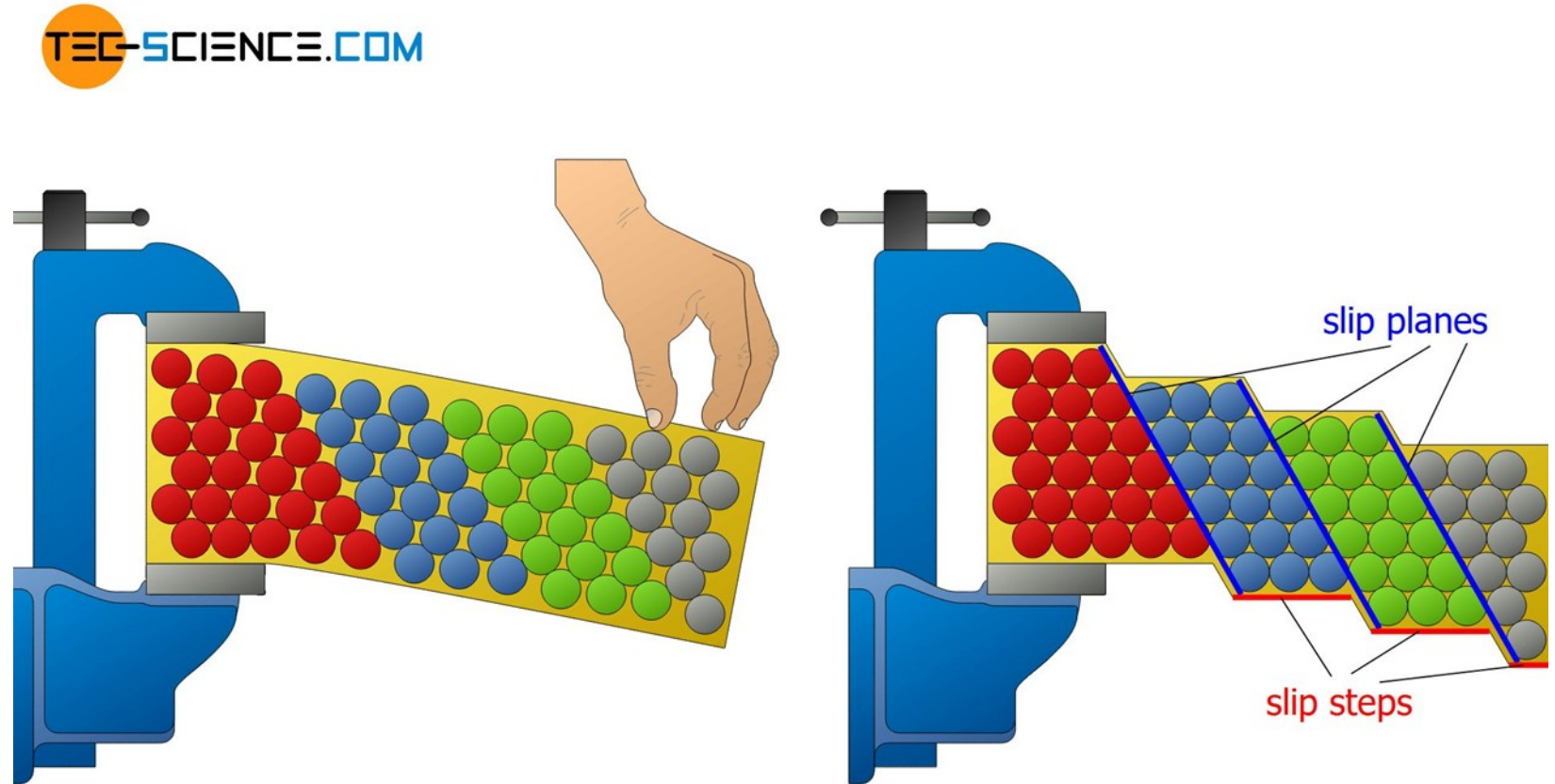
Elastic and Plastic Deformation

Alloys and Heat Treatment

Why do materials return or stay bent?

Main question

- Why does one wire return?
- Why does another stay bent?
- Same force – different result
- Why?



Atomic Crystal Structure of Metals

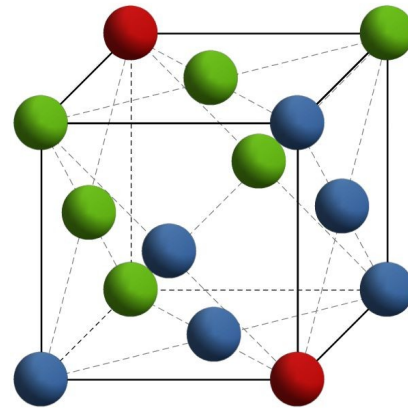
Metals have a regular crystal lattice

Atoms are arranged in an orderly pattern

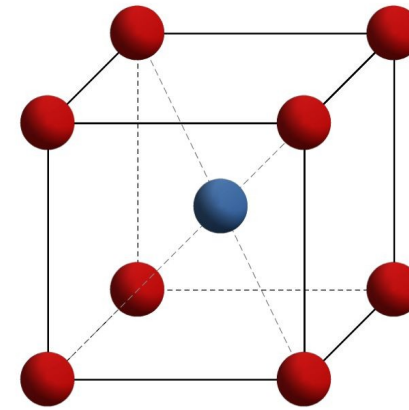
This structure gives metals their special properties

Different metals have different types of crystal structures (example: FCC, BCC)

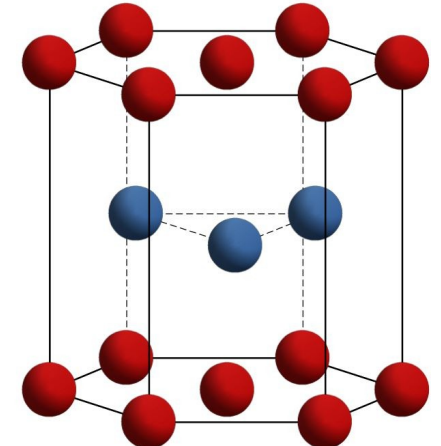
decreasing ductility



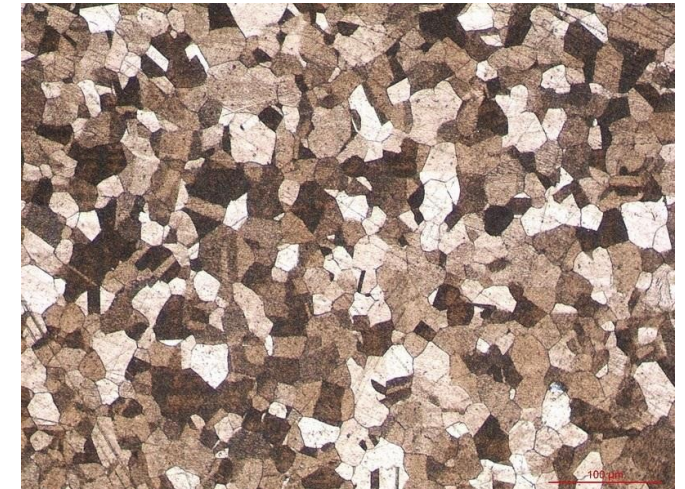
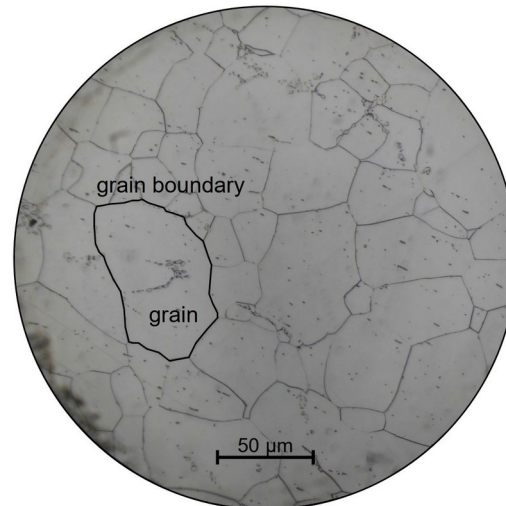
fcc



bcc



hcp



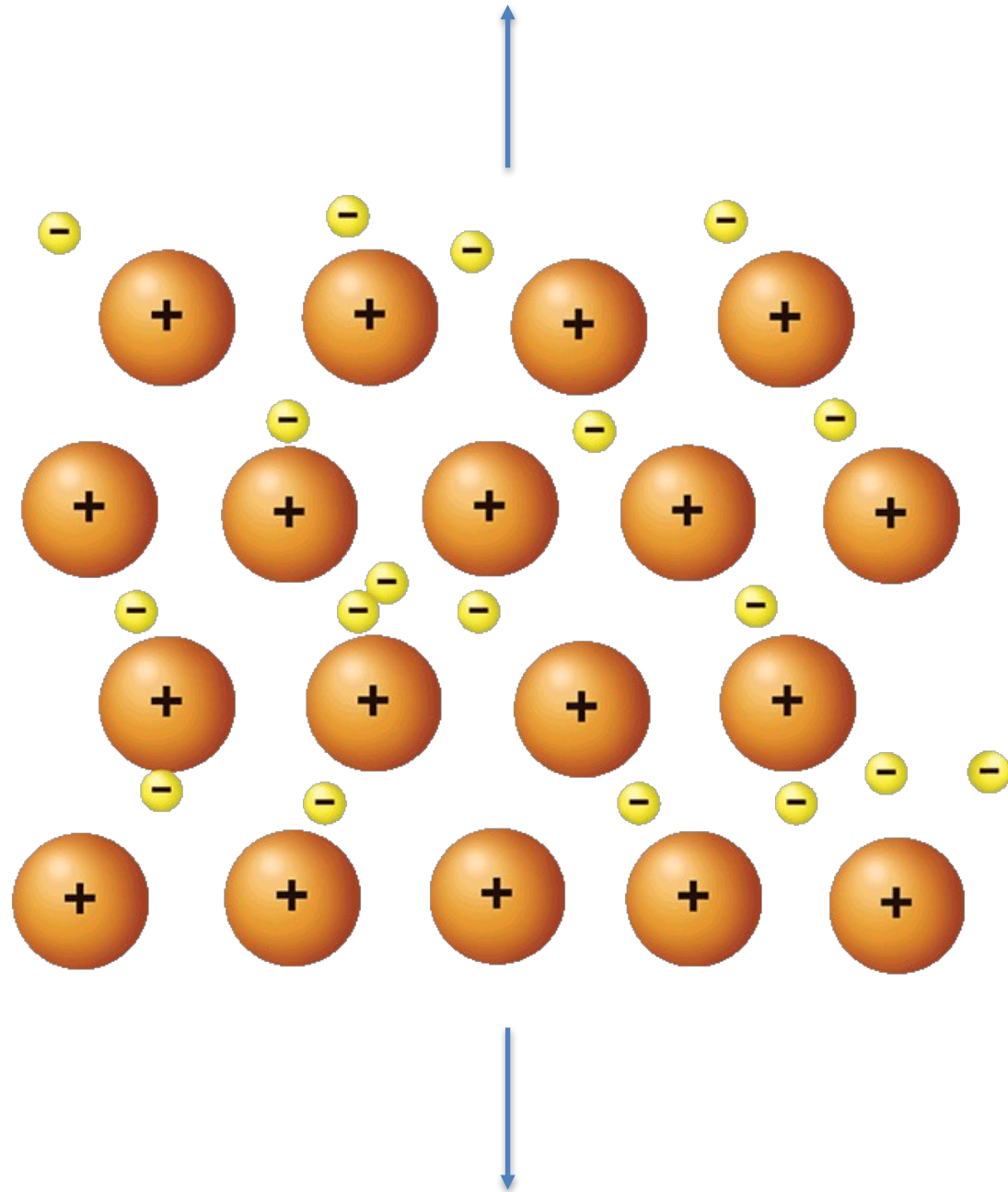
Elastic Deformation

When we apply small force, atoms move a little

They stretch or compress, but stay in their positions

When force is removed, atoms return to original places

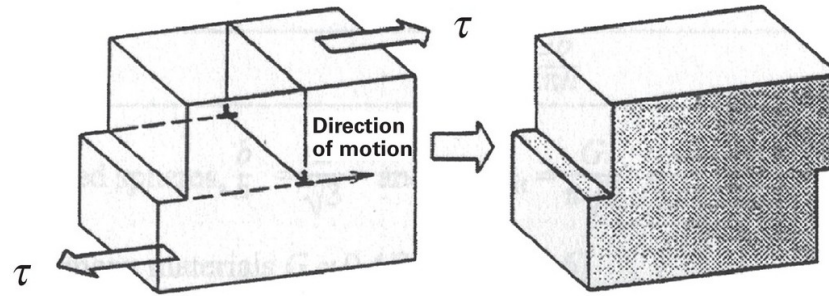
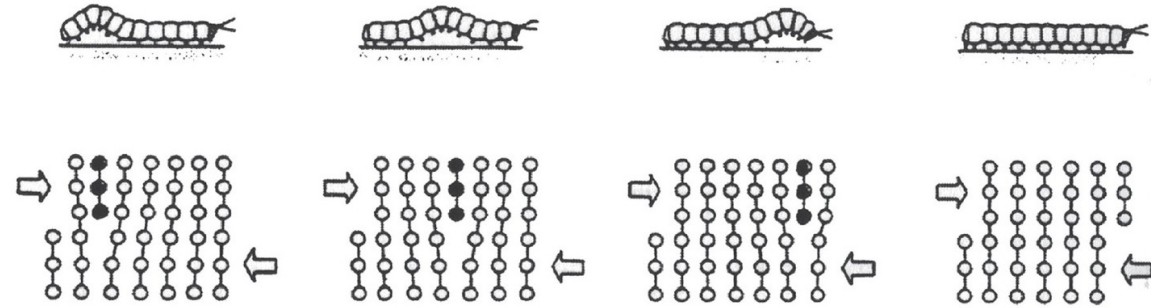
Material returns to its original shape



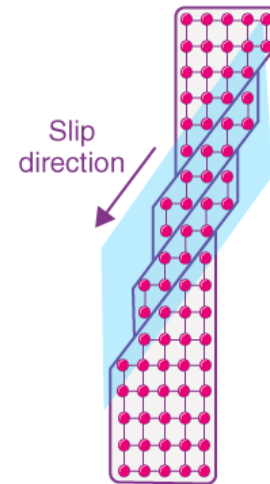
Plastic Deformation

- When force is large, atoms start to slide past each other
- They move along slip planes
- Atoms change their positions permanently
- Material keeps the new shape

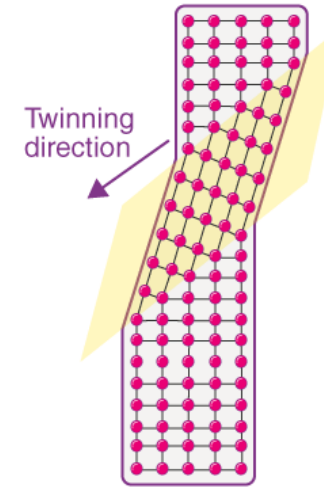
Example: Bent copper wire, aluminum foil, bent spoon



BYJU'S
The Learning App



(a) Slip



(b) Twinning

Comparison: Elastic vs Plastic Deformation

Property

Elastic Deformation

Plastic Deformation

Force

Small

Large

Atom movement

Atoms stretch and return

Atoms slide permanently

Shape after force

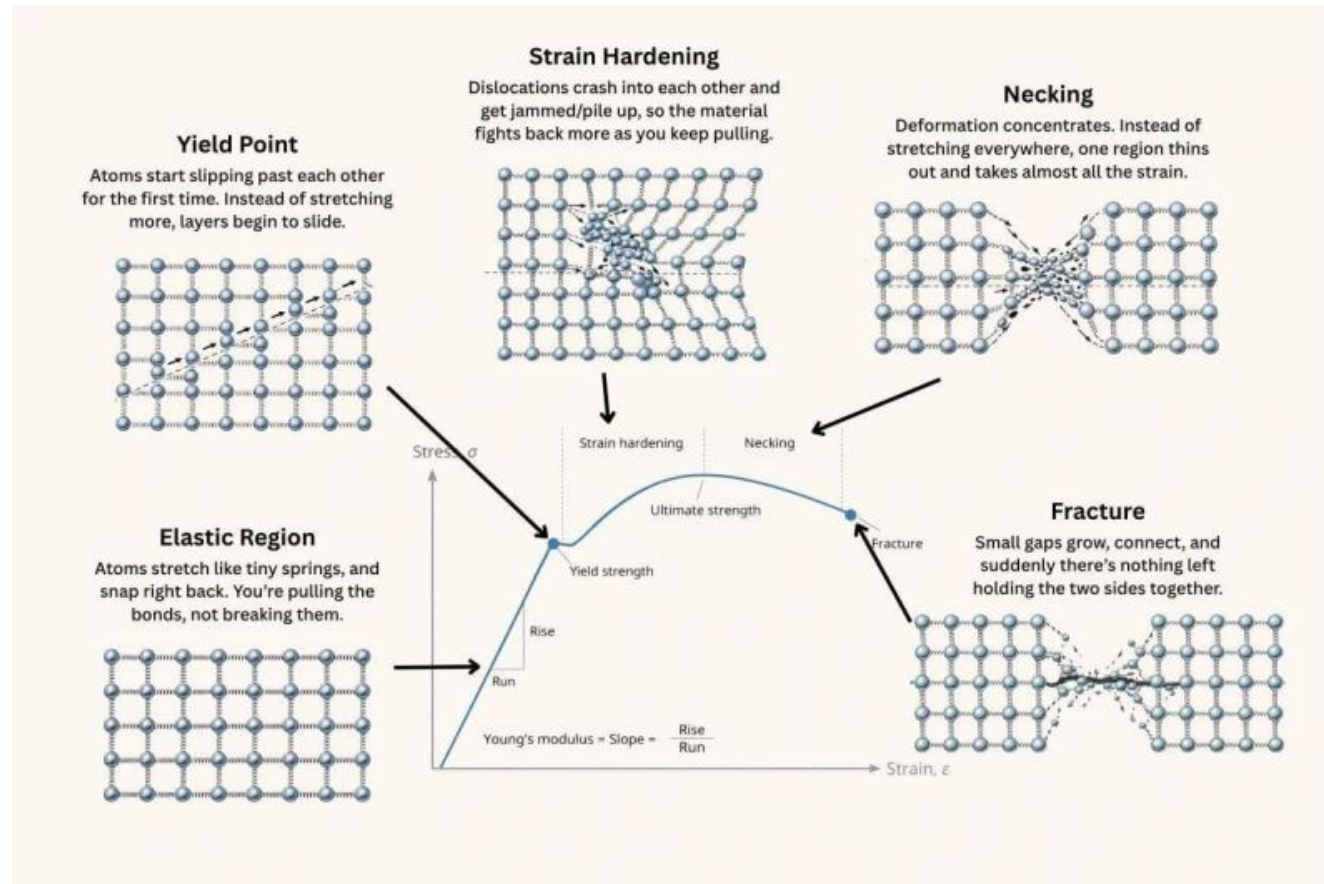
Returns to original

Stays changed

Example

Spring

Bent copper wire



Alloys

- Pure metals are often too soft or weak
- Alloy = mixture of two or more metals
- Alloys are usually stronger and harder than pure metals

Two main types of alloys:

1. Substitution Solid Solution

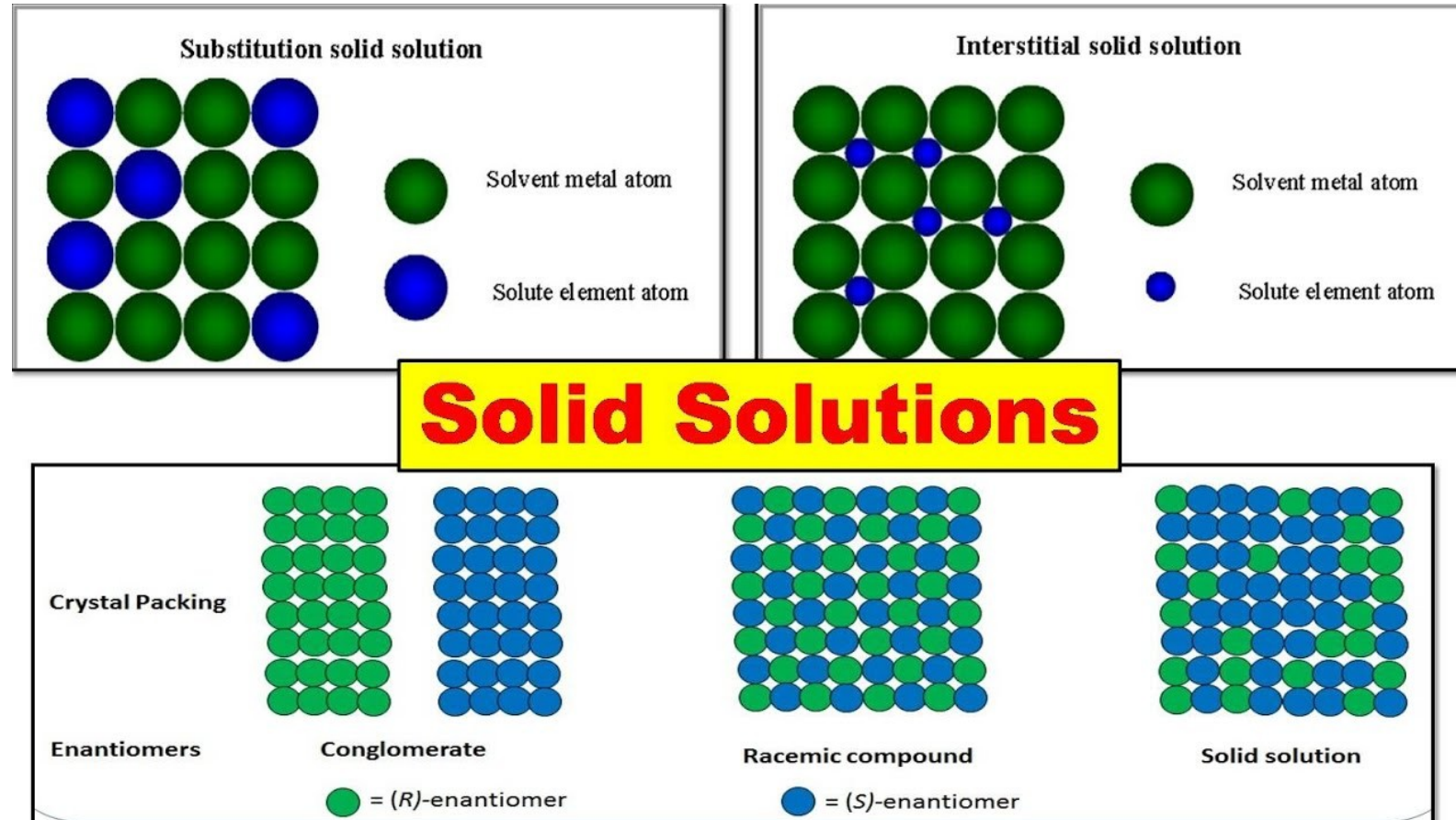
Bigger atoms replace some atoms in the lattice

2. Interstitial Solid Solution

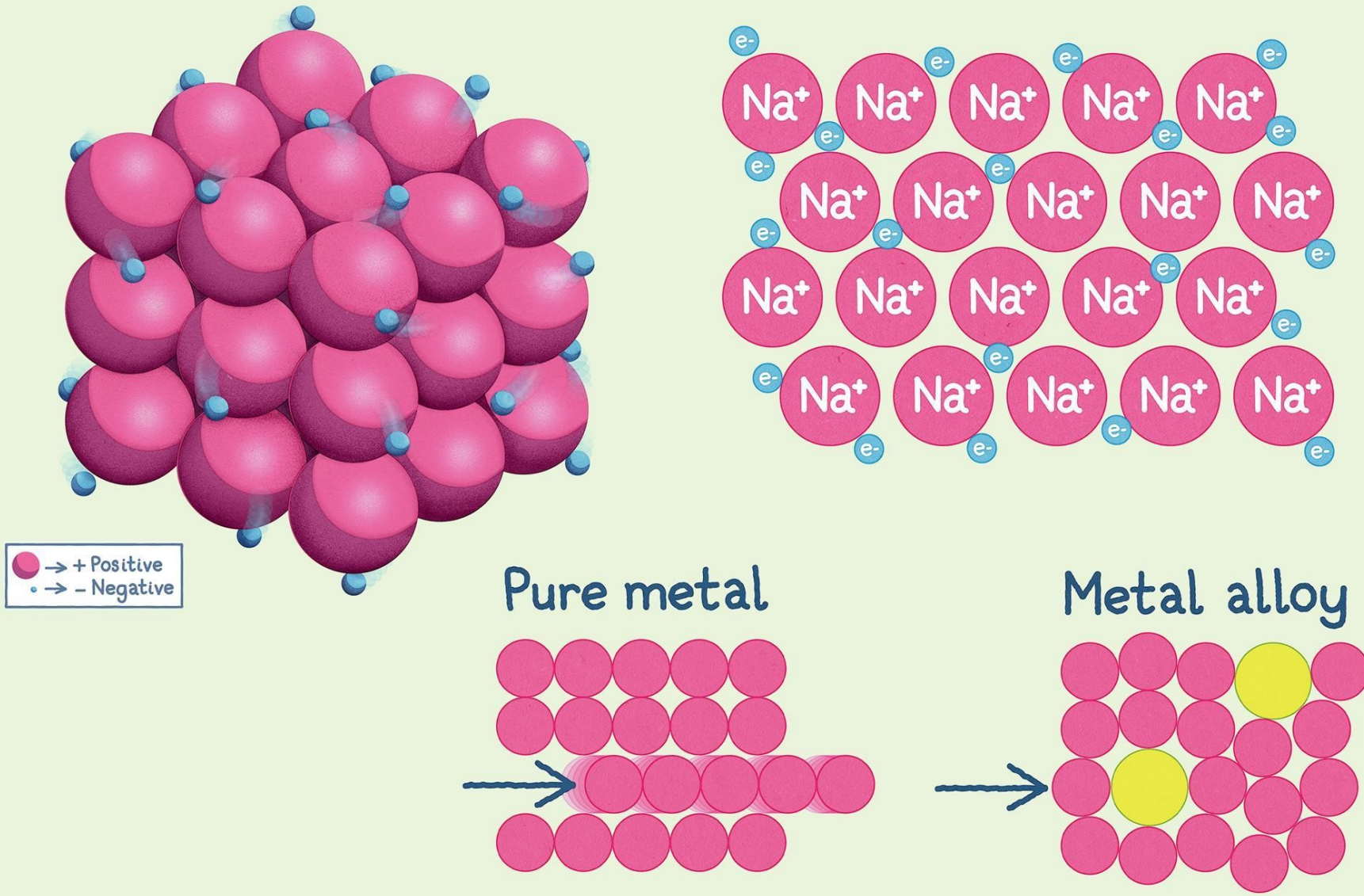
Small atoms fit between the bigger atoms

Example: Bronze = Copper + Tin

Steel = Iron + Carbon



Alloys



That's why alloys stronger than metals

Heat Treatment and Quenching

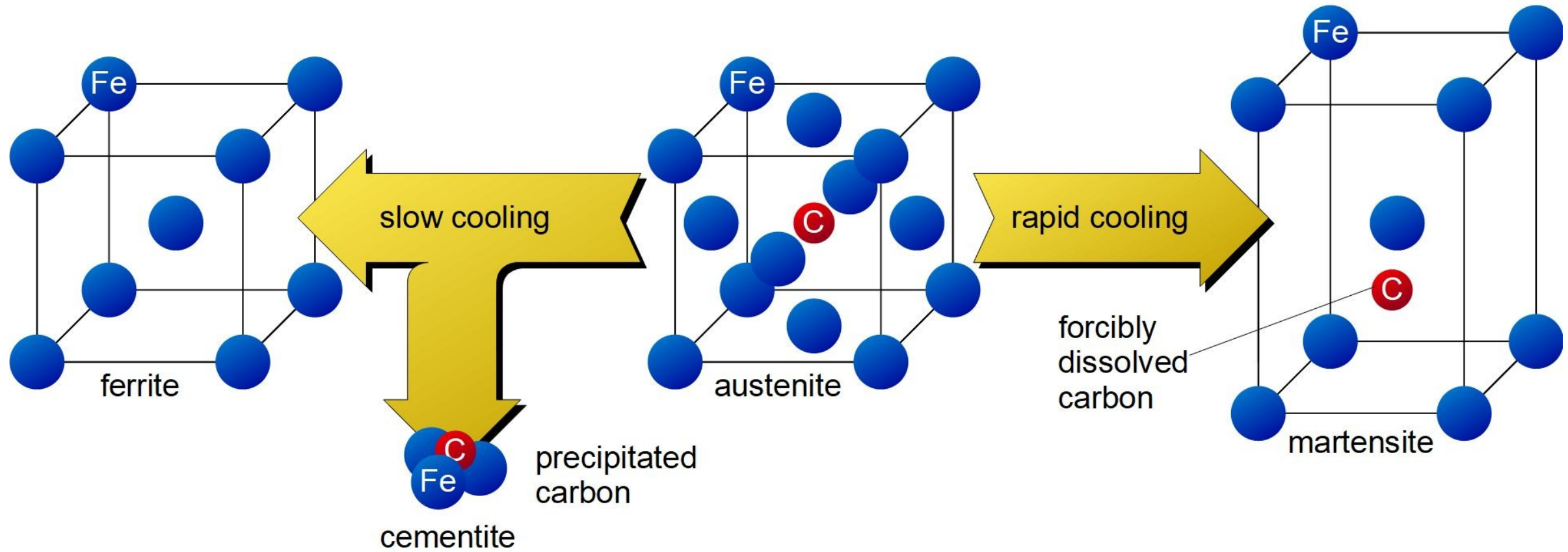
- Heat treatment = controlled heating and cooling of metal
- Quenching = heat metal very hot, then cool it very fast (in water or oil)

Result of quenching:

- Metal becomes much harder and stronger
- But it can also become more brittle

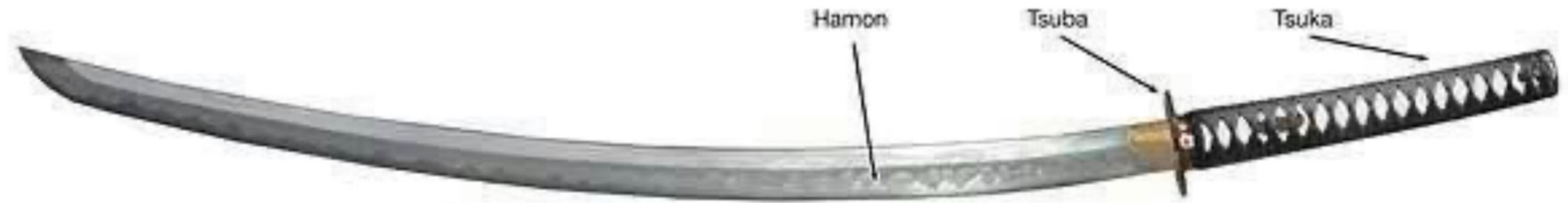


Heat Treatment and Quenching



Japanese Katana – Masterpiece of Materials Science

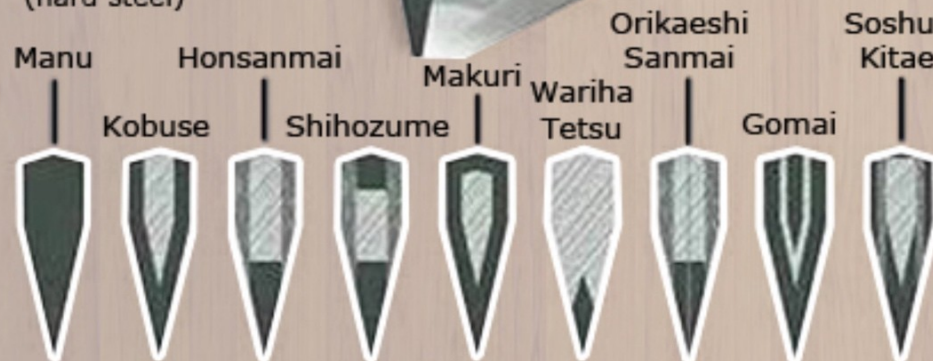
- Japanese sword makers were masters of material science
 - Katana has **multi-layer structure**
 - Hard edge (high carbon, quenched) – very sharp
 - Soft core – flexible and tough
- This combination gives katana both sharpness and strength.



The hamon, or temper wave, demonstrates the strength and the cutting power

Tsuba provides protection for hand and decorative aesthetics

-  **Munetetsu**
(medium steel)
-  **Kawatetsu**
(folded steel)
-  **Shintetsu**
(flexible steel)
-  **Hatetsu**
(hard steel)



Each type of blade represents the different layers of hardness of forged steel. Harder steel form the edge.

The softer steel is in the center to increase flexibility, absorbing the impacts.

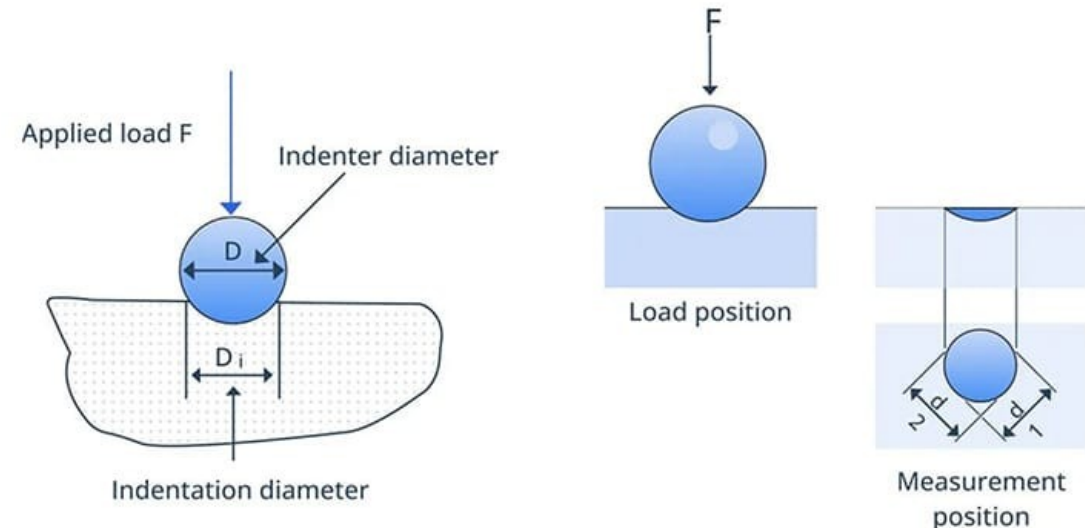
Important properties we need to know:

- **Hardness** – resistance to scratching or indentation
- **Strength** – ability to withstand load without breaking
- **Elasticity** – ability to return to original shape
- **Plasticity** – ability to change shape permanently
- **Brittleness** – breaks without plastic deformation
- **Toughness** – ability to absorb energy before breaking

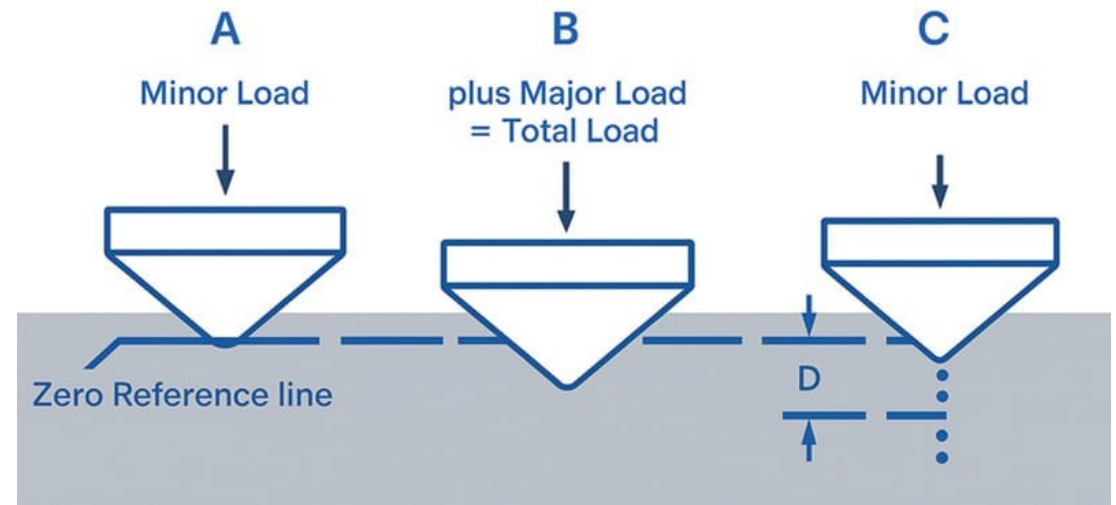
Hardness

- Hardness = how difficult it is to scratch or dent the material
- Measured by Rockwell, Brinell or Vickers tests
- Example: Diamond is the hardest natural material
- In medicine: surgical tools must be hard

Brinell Hardness Test



Rockwell Hardness Test



Strength and Toughness

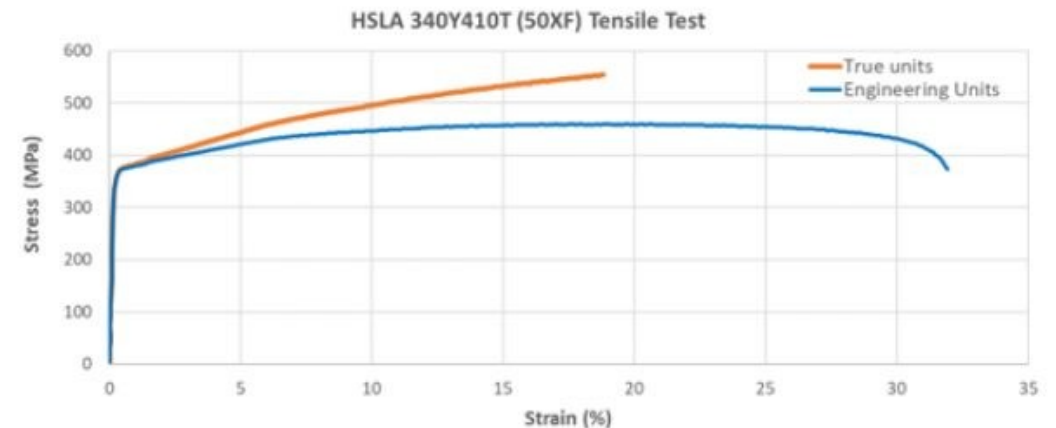
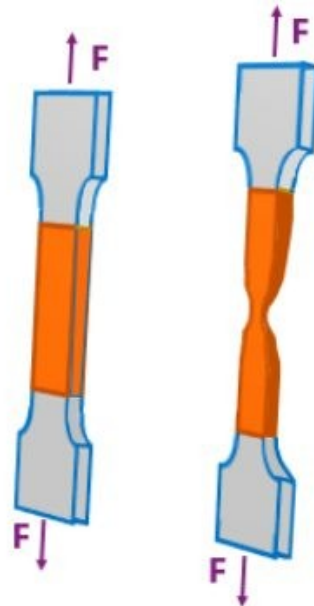
- **Strength** – how much load material can carry
- **Toughness** – how much energy material can absorb before breaking

Example: Good scalpel needs both high strength and good toughness.

True Stress-Strain Curve

True Stress: Applied load divided by instantaneous cross-section area

True Strain: $\ln$ (instantaneous change in length relative to the initial length)



Strength and Toughness

"Here we have two materials.

Material 1: Low-carbon mild steel. It is relatively soft, bends easily, and has good ductility (plasticity).

Material 2: White cast iron. It is very hard, but brittle and almost does not bend.

Question: Which material do you think will **better resist breaking** when we pull it (tensile force)? Which one can withstand a larger force before it finally breaks?

Low-carbon mild steel or white cast iron?



Summary

- Difference between elastic and plastic deformation
- How atoms behave inside metals
- What alloys are and why they are important
- Basics of heat treatment and quenching
- Main mechanical properties of materials (hardness, strength, elasticity, plasticity, toughness)

Understanding these properties helps us choose the right material for medical tools.

Next lecture: Tools and material processing.